Machine Test AI Framework – Zecpath

Objective

Design an AI-driven system that evaluates real-world technical skills through practical machine tests such as coding, debugging, file-based assignments, and system design exercises.

Supported Machine Test Types

Test Type	Description
Coding Problems	Algorithm and logic-based programming tasks
Debugging Tasks	Identify and fix bugs in existing code
File-Based Tasks	Modify or complete project files
Mini System Design	Design scalable software components

High-Level Architecture

```
Candidate Interface (IDE/Web Editor)
            |
            ▼
     Code Capture Engine
            |
            ▼
 Execution Sandbox Engine
            |
            ▼
   Test Case Evaluation
            |
            ▼
 Code Quality Analyzer
            |
            ▼
 Time & Attempt Tracker
            |
            ▼
 Machine Test Scoring Engine
            |
            ▼
 Technical Report Generator
```

Workflow

1. Candidate starts the machine test.
2. Candidate writes or modifies code.
3. System captures code snapshots.
4. Code executes inside a secure sandbox.
5. Test cases are evaluated.
6. Runtime and execution statistics are collected.
7. Code quality is analyzed.
8. Time taken and attempts are recorded.
9. Final score is calculated.
10. Technical report is generated.

Evaluation Parameters

- Correctness
- Runtime Efficiency
- Code Quality
- Problem Solving Ability
- Completion Time

Advantages

- Practical technical assessment
- Objective evaluation
- Automated scoring
- Scalable architecture

Limitations

- Rule-based code quality evaluation
- Limited optimization detection
- No plagiarism detection

Future Improvements

- AI-powered code review
- Plagiarism detection
- Live collaborative coding
- LLM-assisted evaluation